In physics, work is related to force, which is often intuitively defined as a push or a pull on an object. When a force moves and object, we say the force does work on the object. In other words, work can be thought of as the amount of energy it takes to move an object. If F is a constant force that moves an object a distance d, then the *work* W done on the object by the force is defined to be the product of the force times the distance:

$$W = Fd. \tag{1}$$

The SI unit of work is therefore the Newton-meter, also called the *Joule*.

Suppose we have a variable force $F(x)$ that moves an object in the positive direction along the x-axis from point a to point b. To calculate the work done, partition the interval $[a,b]$ and estimate the work done over each subinterval. Let $P = \{x_i\}$ be a partition of $[a,b]$ into equally-spaced subintervals of width $\Delta x = \frac{b-a}{n}$. For each $i = 1,2,\ldots,n$, choose an arbitrary point $x_i^* \in [x_{i-1}, x_i]$. If F is continuous, then for Δx small the force over the ith subinterval is approximately the constant value $F(x_i^*)$, and the work W_i in moving the object over the ith subinterval is then given by Equation (1):

$$W_i \approx F(x_i^*)\Delta x.$$

The total work done over the interval $[a,b]$ is therefore approximately

$$W = \sum_{i=1}^{n} W_i \approx \sum_{i=1}^{n} F(x_i^*)\Delta x.$$

Recognizing this as a Riemann sum, we take the limit as $n \to \infty$ (which exists since F is assumed to be continuous), to define the total work as

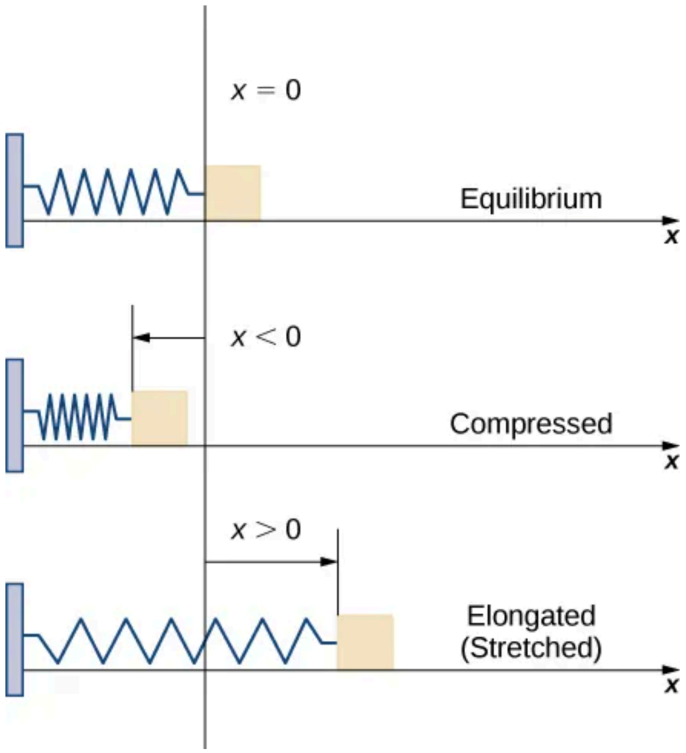
$$\boxed{W := \lim_{n\to\infty} \sum_{i=1}^{n} F(x_i^*)\Delta x = \int_a^b F(x)dx}. \tag{2}$$

1. Show that if F is constant, then Equation (2) reduces to (1).

Name:_____

1 Springs

Consider now the example of the work done to compress or extend a spring. Consider a block attached to a horizontal spring, which moves back and forth as the spring stretches and compresses. For simplicity, assume the block is resting on a frictionless surface. When the spring is at its natural length (at rest), the system is said to be at *equilibrium*. In this state, the spring is neither elongated nor compressed, and in this equilibrium position the block does not move until a force is introduced. Choose coordinates so that the block moves along the x-axis, with $x = 0$ corresponding to the equilibrium position (see below).



According to *Hooke's Law*, the force required to stretch a spring by an amount x from its equilibrium position is

$$F(x) = kx$$

where k is a positive constant, called the *spring constant*, which depends on the physical characteristics of the spring. [1]

[1] In most physics texts, Hooke's Law is stated as the force exerted by the spring on the block, which, by Newton's Third Law, is $F(x) = -kx$.

2. A certain spring has spring constant $k = 120$ N/m. Calculate the work done in stretching this spring 0.1m from its equilibrium position.

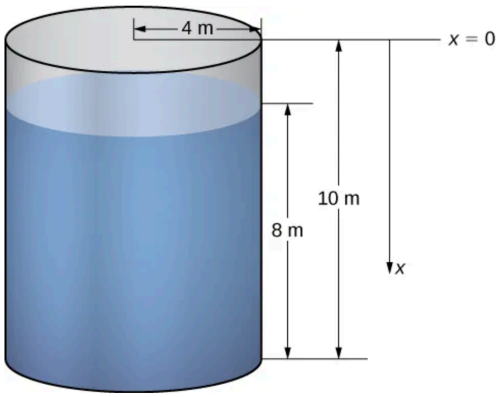
3. Suppose it takes a force of 10 N (in the negative direction) to compress a spring 0.2 m from its equilibrium position. How much work is done in stretching the spring 0.5 m from its equilibrium position?

2 Work Done in Pumping

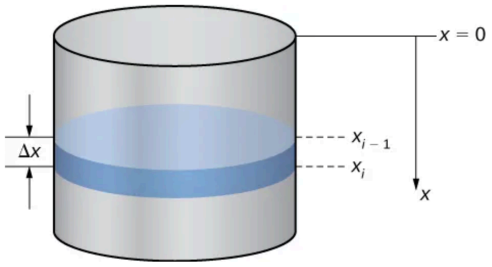
Consider the work done to pump water (or some other liquid) out of a tank. Pumping problems are a little more complicated than spring problems because many of the calculations depend on the shape and size of the tank. In addition, instead of being concerned with the work done to move a single mass, we are looking at the work done to move a volume of water, and it takes more work to move the water from the bottom of the tank than it does to move the water from the top of the tank.

Example 2.1 (Cylindrical tank)**.** Consider a cylindrical tank of radius 4 m and height 10 m which is filled to a depth of 8 m. How much work does it take to pump all the water over the top edge of the tank?

Begin by choosing coordinates such that the x-axis vertically, with the origin at the top of the tank and the downward direction being positive (see below).



In this coordinate system, the water extends from $x = 2$ to $x = 10$. Our strategy will be to partition the interval $[2,10]$ and calculate the work required to lift each individual "layer" of water. Partition $[2,10]$ into n equally-spaced subintervals $[x_{i-1},x_i]$ of length $\Delta x = \frac{10-2}{n} = \frac{8}{n}$. For each $i = 1,2,\ldots,n$, choose an arbitrary point $x_i^* \in [x_{i-1},x_i]$. The figure below shows a representative layer:



In pumping problems, the force required to lift the water to the top of the tank is the force required to overcome gravity, so it is equal to the weight of the water. Given that the density ρ of water is $9800\ \mathrm{N/m^3}$, the force needed to lift each layer is

$$F = \rho V = 9800\pi r^2 \Delta x = 9800\pi(4)^2\Delta x = 156,800\pi\Delta x.$$

We also need to know the distance the water must be lifted. Based on our choice of coordinate system, we can use x_i^* as an approximation of the distance the layer must be lifted. Since the force is constant (that is, independent of x) the approximate work W_i done in lifting the ith layer of water is given by Equation (1):

$$W_i \approx 156,800\pi x_i^* \Delta x.$$

The total work is then approximately

$$W = \sum_{i=1}^{n} W_i \approx \sum_{i=1}^{n} 156,800\pi x_i^* \Delta x.$$

Recognizing this as a Riemann sum, we take the limit as $n \to \infty$ to obtain

$$\begin{aligned} W :&= \lim_{n\to\infty} \sum_{i=1}^{n} 156,800\pi x_i^* \Delta x \\ &= 156,800\pi \int_2^{10} x dx \\ &= 156,800\pi \left[\frac{x^2}{2}\right]_2^{10} \\ &= 7,526,400\pi \text{ J} \\ &\approx 23,644,883 \text{ J}. \end{aligned}$$

For pumping problems, the calculations vary depending on the shape of the tank. The following problem-solving strategy lays out a step-by-step process for solving pumping problems.

PROBLEM-SOLVING STRATEGY

Problem-Solving Strategy: Solving Pumping Problems

1. Sketch a picture of the tank and select an appropriate frame of reference.
2. Calculate the volume of a representative layer of water.
3. Multiply the volume by the weight-density of water to get the force.
4. Calculate the distance the layer of water must be lifted.
5. Multiply the force and distance to get an estimate of the work needed to lift the layer of water.
6. Sum the work required to lift all the layers. This expression is an estimate of the work required to pump out the desired amount of water, and it is in the form of a Riemann sum.
7. Take the limit as $n \to \infty$ and evaluate the resulting integral to get the exact work required to pump out the desired amount of water.

4. Assume a tank in the shape of an inverted cone, with height 12 ft and base radius 4 ft. The tank is full to start with, and water is pumped over the upper edge of the tank until the height of the water remaining in the tank is 4 ft. How much work is required to pump out that amount of water? (In English units, the density ρ of water is 62.4 lb/ft^3.)